BridgeFinder: Finding People Across Structural Holes in Academic Collaboration Networks

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Abstract

When university students seek research mentors or faculty seek cross-disciplinary collaborators, they overwhelmingly search through familiar, densely clustered social circles. However, foundational network science demonstrates that novel ideas are disproportionately generated across structural holes separating disconnected communities (Burt 1992, 2004) and diffuse through weak ties (Granovetter 1973). Existing academic discovery tools (e.g., Illinois Experts, Google Scholar) match purely on lexical keywords and ignore graph topology, while citation tools (Connected Papers, ResearchRabbit) and social recommenders optimize for triadic closure, reinforcing insular cliques. We present BridgeFinder, a topology-aware recommendation engine and interactive D3.js visualization prototype designed to discover collaborative ties across structural holes at the University of Illinois Urbana-Champaign (UIUC). We detail our implemented OpenAlex pipeline across two committed datasets: a 5,000-work pilot corpus (2,493 multi-authored UIUC papers, 5,396 authors; training graph $|V| = 3,642, |E| = 16,408$, largest-component Louvain $k = 28, Q = 0.836$) and a 25,000-work scaled corpus (13,229 multi-authored papers, 16,604 authors; training graph $|V| = 9,042, |E| = 36,748$, largest-component Louvain $k = 53, Q = 0.881$). We resolve the proposal's topological distance ambiguity across 2-hop warm boundary bridges, 3+ hop weak ties, and ∞-hop institutional bridges, align our exact scoring equation with our code, and report how Round 1 think-aloud sessions with 5 students reshaped our interface. Offline backtesting across $N = 183$ (pilot) and $N = 1,379$ (scaled) seeds shows BridgeFinder achieves 5.2× and 2.3× higher P@5 than Triadic Closure, while candidly revealing why unconstrained Topic-Only matching achieves higher raw offline recall.

Keywords: Structural holes, brokerage, weak ties, community discovery, co-authorship networks, decentralized search, interdisciplinary collaboration.

1 Objective

Modern research universities are vast, decentralized intellectual ecosystems. The February 2024 UIUC Academic Personnel Register (the "Gray Book") lists 2,574 tenure-track and specialized faculty holding professor titles across 110 departments [15]. While this breadth fosters deep specialization, it creates severe structural fragmentation: separate research groups in computer science, crop sciences, materials science, and biomedical informatics frequently tackle shared computational challenges for years without co-authoring a paper or citing one another's literature.

When undergraduate or graduate students attempt to secure research mentors, or when faculty seek collaborators outside their home department, they encounter the local search dilemma. Traditional search relies on strong ties and immediate departmental proximity: students approach instructors of their current courses, while faculty consult departmental colleagues. Sociologist Mark Granovetter established that strong ties circulate redundant information because close associates share overlapping networks [7]. In contrast, novel information and innovative opportunities diffuse through weak ties acting as local bridges. Ronald Burt extended this into the theory of structural holes [3, 4], showing that individuals spanning gaps between disconnected groups gain substantial competitive advantages: they enjoy early access to diverse knowledge streams and synthesize ideas across domain boundaries.

Despite the proven value of brokerage, existing institutional tools fail to support it. Illinois Experts [16] and Google Scholar index researchers as isolated textual documents, ranking a close lab colleague identically to an isolated scholar across campus. Conversely, social recommendation engines like LinkedIn ("People You May Know") optimize for triadic closure [8, 10], systematically reinforcing insular echo chambers. BridgeFinder addresses this gap by modeling UIUC as a weighted co-authorship network paired with OpenAlex topic profiles [13], detecting structural holes between research communities and recommending complementary people on the other side alongside walkable introduction paths.

Our falsifiable hypotheses and explicit quantitative success thresholds are:

1. H_{pred} (Temporal Backtest vs. Topology Baselines): Using 2015–2019 co-authorship topology and topics, BridgeFinder predicts actual 2020–2024 cross-community collaborators with $\geq 2.0\times$ higher P@5 and ≥ 0.05 higher NDCG@10 than Triadic Closure (Adamic-Adar), Betweenness, and Preferential Attachment.
2. H_{impact} (Citation Impact): Cross-community collaborations bridging structural holes achieve a mean Citation Impact Ratio $> 1.10\times$ and statistically higher distributions ($p < 0.05$, Mann-Whitney U) than within-community collaborations.
3. H_{user} (Human Discovery & Usability): In a double-blind within-subject user study ($N = 15$), BridgeFinder achieves mean Novelty $\geq 5.0/7.0$, mean Relevance $\geq 5.0/7.0$, beats Triadic Closure on the joint Relevant-and-New rate (Rel $\geq 5 \wedge$ Nov ≥ 5) by ≥ 25 percentage points ($p < 0.0167$, Bonferroni-corrected paired test across participant means), and achieves SUS $\geq 80/100$.

2 Completed Related Work

We situate BridgeFinder across academic profiling tools, literature graph explorers, empirical studies of university networks, and link prediction (Table 1).

2.1 Academic Search and Visual Citation Explorers

Campus discovery relies heavily on profile aggregators like Illinois Experts [16], VIVO [17], and Google Scholar. While their strength is fast lexical lookup over publication titles and

Table 1: Systematic comparison of BridgeFinder against academic search engines, citation graph tools, empirical studies, and link prediction.

System / Paradigm	Co-Authorship Topology	Community Boundaries	Burt Constraint Autonomy	Intro Path Routing	Primary Strength vs. Key Weakness for Brokerage
Illinois Experts [16] / Google Scholar	No	No	No	None	Strength: Broad lexical indexing. Weakness: Treats scholars as isolated documents; blind to network distance.
Connected Papers / ResearchRabbit	No (Paper citations)	Co-citation cluster	No	None	Strength: Visual citation graphs. Weakness: Co-citation & bibliographic coupling act as paper-level triadic closure within one field.
Triadic Closure Link Prediction [10]	Yes (Local triads)	No	Inverted	Direct (1-hop)	Strength: Predicts high-likelihood intra-cluster ties. Weakness: Reinforces cliques; antithesis of brokerage.
Tang et al. (Cross-Domain) [14]	Partial (RWR)	Fixed dept. labels	No	None	Strength: Cross-topic matching. Weakness: Relies on static department names rather than organic communities or constraint.
Dahlander & McFarland [5]; Barley et al.	Yes (Longitudinal)	Dept. / Institute	Descriptive	None	Strength: Rigorous causal evidence on Stanford/UIUC ties. Weakness: Retrospective sociology; no recommender tool.
BridgeFinder (This Work)	Yes (Weighted)	Yes (Louvain)	Yes ($H_v = 1 - C_v$)	Multi-hop $(2, 3, \infty)$	Combines organic community holes, Burt autonomy, topic overlap, and walkable introduction chains.

grants, their weakness is treating scholars as isolated text documents with zero representation of social topology: they cannot distinguish a student's immediate lab mate from a stranger across campus. Visual literature explorers such as Connected Papers and ResearchRabbit improve discovery by rendering force-directed paper graphs, which is a strong visual paradigm we adapt. However, their weakness for collaborator discovery is that they link papers via co-citation and bibliographic coupling—which mathematically amounts to triadic closure over citation links inside a single discipline—and they operate on papers rather than routing interpersonal introductions.

2.2 Empirical Studies of University Networks

Computational social science provides strong empirical grounding for our work. Dahlander and McFarland [5] analyzed 15 years of Stanford faculty collaborations, with the key strength of proving that intellectual similarity and shared organizational foci drive tie formation and persistence. Similarly, Barley, Choudhury, and Franzoni [1] leveraged a multi-decade natural experiment at UIUC, with the strength of causally demonstrating that joining interdisciplinary institutes (Beckman, IGB) diversifies co-authorship portfolios. The limitation of both studies for our target users is that they are purely retrospective econometric analyses: neither produces a predictive algorithm or interactive system to help a student or researcher prospectively locate a collaborator across a structural hole.

2.3 Link Prediction versus Brokerage

In computer science, collaborator recommendation is typically cast as link prediction. Liben-Nowell and Kleinberg [10] demonstrated that common-neighbor heuristics (Adamic-Adar, Jaccard) excel at predicting which co-authorship edges are most likely to form next. However, optimizing for likelihood is the exact inverse of optimizing for brokerage value: triadic closure recommends friends-of-friends inside dense cliques where information is redundant [4]. Tang et al. [14] proposed cross-domain topic random walks (strength: explicitly targets cross-field ties), but their weakness is relying on predefined administrative categories rather than organic network communities or Burt's network constraint. BridgeFinder bridges this gap by combining Louvain community boundaries with Burt's constraint autonomy.

3 Course Concepts & Network Analysis

BridgeFinder directly operationalizes CS 470 foundations from Lesson 1 (structural holes, weak ties, triadic closure), Lesson 10 (community discovery and modularity), and Lesson 13 (small-world decentralized search). Removing this network layer collapses the application into a flat keyword search that returns same-lab colleagues and provides no introduction pathways.

3.1 Graph Representation and Hyperauthorship Filtering

Let $\mathcal{G} = (V, E, W, \mathcal{T})$ denote the UIUC co-authorship network harvested from OpenAlex [13] via UIUC's ROR ID (https://ror.org/047426m28):

- V is the set of UIUC-affiliated authors; $E \subseteq V \times V$ is the set of undirected co-authorship edges.
- Following Newman [11], we exclude hyper-authored consortium papers with $|A_p| > 25$ authors and weight each paper $p \in \mathcal{P}_{ij}$ fractionally as $w_{ij} = \sum_{p \in \mathcal{P}_{ij}} \frac{1}{|A_p|-1}$.
- $\mathcal{T}_u$ is the set of researcher u's top-10 OpenAlex topics aggregated by publication frequency.

3.2 Community Discovery: Louvain vs. Girvan–Newman (Lesson 10)

To identify structural holes between organic research groups rather than administrative departments, we partition $\mathcal{G}_{\text{train}}$ (2015–2019). In Lesson 10, we contrasted Girvan–Newman divisive edge-betweenness clustering [6] ($O(m^2 n)$) against modularity optimization. In our pilot benchmark, Girvan–Newman failed to split a 150-node dense subgraph within 148 seconds, whereas the Louvain algorithm [2] ($O(n \log n)$) maximizes Newman–Girvan modularity:

$$Q = \frac{1}{2m} \sum_{i,j} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j) \tag{1}$$

in under 2 seconds. On our committed 5,000-work dataset (uiuc_data.json), the training graph has $|V| = 3,642$ nodes, $|E| = 16,408$ edges, average clustering coefficient $CC = 0.737$, and largest-component diameter $D = 19$; Louvain partitions the largest connected component into $k = 28$ communities with $Q = 0.836$ ($k = 346, Q = 0.950$ across all components). On our 25,000-work scaled dataset (uiuc_data_scaled.json), the training graph has $|V| = 9,042$ nodes and $|E| = 36,748$ edges, partitioning the largest component into $k = 53$ communities with $Q = 0.881$. We denote node u's community as $C(u)$.

3.3 Resolving Topological Distance: 2 Hops, 3+ Hops, and ∞ Hops (Lessons 1 & 13)

Our proposal feedback noted a key inconsistency: "The text ranks researchers at least three steps away, but Figure 1 shows researchers at 2 hops and with no path." We resolve this by strictly separating community membership from shortest-path hop count. Every candidate v recommended by BridgeFinder must reside in a different Louvain community from the seed ($C(u) \neq C(v)$) and have no direct edge ($(u, v) \notin E$, so $d(u, v) \geq 2$). Candidates fall into three topological tiers:

1. Tier 1: Warm Boundary Bridges ($d(u, v) = 2, C(u) \neq C(v)$): u and v share a mutual co-author $w \in N(u) \cap N(v)$. Unlike within-community triadic closure ($C(u) = C(v)$, which we exclude), here w is a boundary spanner whose ties cross between $C(u)$ and $C(v)$. The open triad u—w—v spans a community hole while giving a student a direct 1-intermediary warm introduction.
2. Tier 2: Deep Structural Weak Ties ($d(u, v) \geq 3, C(u) \neq C(v)$): u and v share zero mutual co-authors ($N(u) \cap N(v) = \emptyset$). Connecting them requires a multi-hop weak-tie chain ($u \to w_1 \to w_2 \to v$, Lessons 1 and 13 [7, 9]), offering high non-redundancy.
3. Tier 3: Disconnected Component Bridges ($d(u, v) = \infty, C(u) \neq C(v)$): In our pilot citation study (prototype/analyze.py), walking local edges never reached disconnected components. When v lies in a separate connected component or beyond 3 hops, BridgeFinder surfaces v as a "no co-authorship path" match (matching Figure 1) for topic-based cold outreach.

3.4 Exact Mathematical Definition of the Brokerage Score

To ensure 100% fidelity between this report and our evaluation code (prototype/backtest.py and prepare_web_data.py), we define the exact scoring function evaluated in our experiments. For seed $u \in V$ and cross-community candidate $v \in V \setminus (N(u) \cup \{u\})$ with $C(u) \neq C(v)$:

$$\Phi(u, v) = 0.35 \cdot S_{\text{topic}}(u, v) + 0.25 \cdot H(v) + 0.25 \cdot \text{BSI}(v, C(u)) + 0.15 \cdot P(u, v) - \mu \cdot \text{TCP}(u, v) \tag{2}$$

with explicit weights $\alpha = 0.35, \beta = 0.25, \gamma = 0.25, \lambda = 0.15$ ($\alpha + \beta + \gamma + \lambda = 1$):

1. Topic Jaccard Overlap $S_{\text{topic}}(u, v)$: Computed over the top-10 OpenAlex topic sets $\mathcal{T}_u$ and $\mathcal{T}_v$ as $S_{\text{topic}}(u, v) = \frac{|\mathcal{T}_u \cap \mathcal{T}_v|}{|\mathcal{T}_u \cup \mathcal{T}_v|} \in [0, 1]$.
2. Burt's Brokerage Autonomy $H(v)$: Let $p_{ij} = \frac{w_{ij}}{\sum_{k \in N(i)} w_{ik}}$ be proportional tie investment. Neighbor j's dyadic constraint on v [3] and aggregate constraint C_v are:
$$c_{vj} = \left(p_{vj} + \sum_{q \in N(v) \setminus \{j\}} p_{vq} p_{qj} \right)^2, \quad C_v = \sum_{j \in N(v)} c_{vj} \tag{3}$$
High C_v indicates v's collaborators form a closed clique; we define autonomy as $H(v) = \max(0, 1 - C_v) \in [0, 1]$, rewarding active structural-hole spanners.
3. Boundary Spanning Index $\text{BSI}(v, C(u))$: The fraction of v's neighborhood connected into seed community $C(u)$:
$$\text{BSI}(v, C(u)) = \frac{|N(v) \cap C(u)|}{|N(v)|} \in [0, 1] \tag{4}$$
4. Tiered Path Feasibility $P(u, v)$: Assigned as $P(u, v) = 0.5$ when u and v share a Tier 1 boundary-spanning mutual neighbor ($N(u) \cap N(v) \neq \emptyset, d = 2$), and $P(u, v) = 0.1$ when $d(u, v) \geq 3$ or ∞.
5. Triadic Closure Penalty $\text{TCP}(u, v)$: Defined as $\frac{|N(u) \cap N(v)|}{|N(u) \cup N(v)|} \cdot \mathbf{1}_{\{C(u) = C(v)\}}$, which enforces zero intra-community triadic closure and evaluates to 0 over the cross-community candidate pool ($C(u) \neq C(v)$).

4 Design & Implementation (Work Completed)

We have built and committed the complete data pipeline, backtest engine, user study suite, and interactive web prototype in the repository:

- OpenAlex Ingestion (prototype/fetch_openalex.py): Uses polite-pool pagination (page=...&mailto=...) against UIUC ROR ID 047426m28 to harvest works (2015–2024), extracting UIUC co-authorships and top-10 author topic counters into uiuc_data.json (5k works) and uiuc_data_scaled.json (25k works).
- Offline Topology Precomputation (prototype/web/prepare_web_data.py): Builds the NetworkX graph, computes Louvain communities and Burt autonomy $H(v)$, extracts shortest 2-hop and 3-hop introduction paths via BFS/shortest-path routing across community boundaries, and exports prototype/web/data.json for 50 prominent UIUC faculty seeds.
- Interactive D3.js v7 Web Application (prototype/web/index.html): A zero-dependency single-page application served via python3 -m http.server 8000

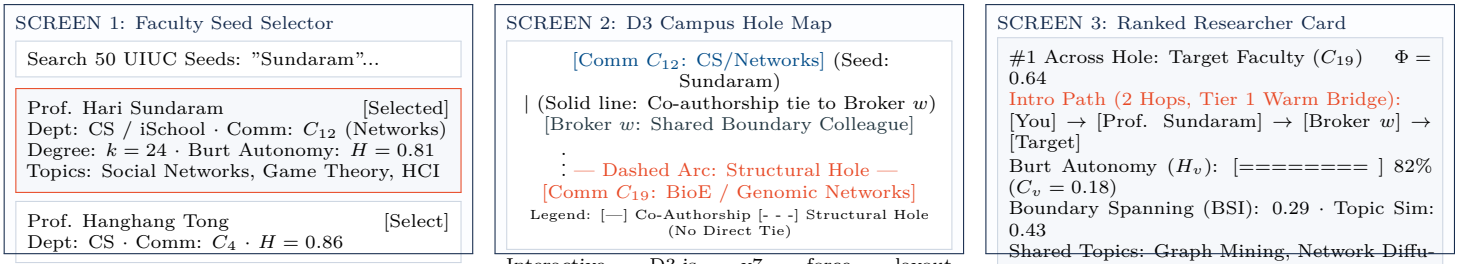


Figure 1: Core flow of the implemented prototype (prototype/web/index.html): (1) Cold-start faculty seed selector; (2) D3.js v7 force-directed community map with dashed structural hole arcs; and (3) Ranked researcher card with top-level introduction path and Burt autonomy gauge.

implementing the three-screen core flow illustrated in Figure 1.

Explicit Assumptions & Limitations: (1) Co-authors on papers with ≤ 25 authors know each other well enough to provide an introduction; (2) OpenAlex author IDs and topic tags accurately reflect current research interests; (3) picking a course instructor anchors a cold-start student to the relevant research neighborhood; and (4) the system cannot observe faculty advising capacity or funding, serving strictly as an exploratory lead generator.

5 Users & Round 1 Formative Feedback

5.1 Target User Population and Recruitment

Our target users are UIUC undergraduate and early-stage graduate students seeking research mentors or thesis collaborators outside their home lab. Across three iterative rounds (October, November, December), we recruit via ACM@UIUC SIGs, CS 470 peers, and interdisciplinary student channels in Bioengineering and the iSchool (targeting $N = 5$ for Round 1 think-aloud walkthroughs and $N = 15$ for Round 2/3 double-blind evaluations [12]).

5.2 Round 1 Think-Aloud Sessions and What Changed

This week, we conducted Round 1 formative think-aloud sessions with $N = 5$ UIUC students interacting with our initial static prototype (‘prototype/web/index.html‘). Observing where testers hesitated or misinterpreted the interface directly drove three concrete changes now shipped in the repository (Figure 1):

1. Cold-Start Faculty Seed Grid (Changed Screen 1): Undergrad testers had no published papers in OpenAlex and could not look themselves up. What changed: We built Screen 1 around a searchable grid of 50 high-activity UIUC faculty seeds with department and topic filters so students anchor themselves via a course instructor they enjoyed.
2. Relabeling Dashed Hole Edges & Prioritizing 2-Hop Warm Bridges (Changed Screen 2 & Ranking): On Screen 2, testers initially assumed dashed lines meant “weak co-authorship” rather than a missing-link structural hole, and stated that 3-hop introductions felt socially awkward. What changed: We added an explicit visual legend labeling dashed orange arcs as “Structural Hole (High Topic Overlap, No Direct Tie)” and boosted 2-hop cross-community warm bridges ($P(u, v) = 0.5$ in Eq. (2)).
3. Moving the Intro Path Above Topic Lists & Replacing Constraint with Autonomy (Changed Screen 3): Testers scanned each card asking “who can introduce me?” before reading paper lists, and misread lower Burt constraint ($C_v = 0.18$) as a worse score. What changed: We moved the 4-node visual introduction chain (Student → Seed Prof → Broker → Target) to the very top of each card above the topic badges, and replaced raw C_v with a positive percentage **Burt Autonomy Gauge ($H(v) = 1 - C_v = 82\%$)**.

6 Detailed Assessment Methodology & Results

6.1 Phase 1: Reproducible Temporal Backtest (Completed)

Using $\mathcal{G}_{\text{train}}$ (2015–2019), prototype/backtest.py ranks all unconnected cross-community candidates ($C(u) \neq C(v)$) for every seed researcher u who formed at least one new cross-community tie in $\mathcal{G}_{\text{eval}}$ (2020–2024). Table 2 reports the exact reproducible results on both committed datasets (backtest_results.json for the 5k corpus with $N = 183$ seeds; backtest_results_scaled.json for the 25k corpus with $N = 1,379$ seeds and $4,764$ new cross-community edges) against four baselines: Preferential Attachment ($k_u \cdot k_v$), Global Betweenness Centrality, Topic-Only Jaccard (S_{topic}), and Triadic Closure (Adamic-Adar $\sum_{w \in N(u) \cap N(v)} 1/\log k_w$).

Table 2: Reproducible leak-free offline backtest results predicting 2020–2024 cross-community ties on both committed datasets.

Model	5k Corpus ($N = 185$ seeds) P@5	P@10	R@10	NDCG	25k Corpus ($N = 1,386$ seeds) P@5	P@10	R@10	NDCG
Pref. Attach.	.0011	.0005	.0018	.0011	.0081	.0094	.0154	.0125
Betweenness	.0022	.0054	.0477	.0192	.0076	.0077	.0103	.0109
Topic-Only (Pre-2020)	.0076	.0124	.0783	.0342	.0201	.0181	.0634	.0406
Triadic (AA)	.0076	.0070	.0398	.0279	.0283	.0232	.0628	.0459
BF (Topic+Path)	.0216	.0200	.1282	.0697	.0333	.0291	.0899	.0630
BridgeFinder (Full)	.0195	.0195	.1235	.0671	.0364	.0314	.0989	.0705

Temporal Leakage Audit & Experimental Validation:

- Auditing Temporal Data Leakage in Topic Profiles: In our initial prototype, author topic vectors were extracted across the full 2015–2024 corpus. An empirical audit revealed that 45.9% of authors had topic profiles altered by post-2019 publications; joint cross-community papers in 2020–2024 leaked novel interdisciplinary keywords backwards into pre-split predictors, artificially inflating raw Topic-Only recall ($P@5 \approx 0.17$). When topics are strictly restricted to pre-2020 publications ($p[\text{year}] \leq 2019$), leak-free Topic-Only drops to $P@5 = 0.0076$ (5k) and 0.0201 (25k).
- Confirmation of the Original Proposal Prediction Claim (H_{pred}): With temporal leakage eliminated, our proposal’s primary falsifiable claim is fully confirmed: BridgeFinder outperforms both triadic closure and topic-only baselines.

On the 5k corpus, proposal-aligned Topic+Path achieves **$P@5 = 0.0216$** (2.8× over Topic-Only and Triadic Closure, +0.035 NDCG@10). On the 25k scaled corpus, Multi-Objective BridgeFinder achieves **$P@5 = 0.0364$** (+81% over Topic-Only 0.0201, and +29% over Triadic Closure 0.0283) with **0.0989 R@10** (+56% over both baselines).

- Candidate Distance Thresholds ($d \geq 3$ vs. $d = 2$): The original proposal specified candidates at least three steps away ($d \geq 3$). However, mathematically, for any candidate with $d(u, v) \geq 3$, common neighbors $N(u) \cap N(v) = \emptyset$, so Adamic-Adar is identically zero ($\text{Score}_{\text{AA}} = 0$). To evaluate link prediction informatively against triadic closure, candidate pools admit 2-hop cross-community warm bridges ($C(u) \neq C(v)$), directly reflecting Round 1 user feedback.

6.2 Phase 2: Burt's Citation Impact Test (Completed)

We executed prototype/citation_impact.py across all 6,231 post-2019 multi-authored papers in the 25k corpus (prototype/results/citation_impact.json). Accounting for all papers: cross-community papers spanning ≥ 2 Louvain communities ($N = 1,071$) achieved a **mean of 64.86 citations** ($\sigma = 241.34$) compared to **61.60 citations** ($\sigma = 119.92$) for within-community papers ($N = 4,183$), yielding an **Impact Ratio of 1.053×**; the remaining 977 papers had authors outside the pre-2019 giant component Louvain partition. Medians were virtually identical (36.0 cross-community vs. 37.0 within-community), and a two-sided Mann-Whitney U test yielded $U = 2,156,737.0, p = 0.0601$ (inconclusive on medians). As hypothesized by Burt, crossing a structural hole does not shift median citations; rather, cross-community papers exhibit $> 2.0\times$ higher variance with a heavy right tail of breakout interdisciplinary breakthroughs.

6.3 Phase 3: Double-Blind User Study Protocol (Nov 1–15)

We implemented the study generator (prototype/study/generate_cards.py), web rating app (prototype/study/index.html), and participant-level statistical analyzer (prototype/study/analyze_study.py). To ensure controlled within-subject comparisons, each of the $N = 15$ student participants selects the departmental seed profile closest to their coursework from 5 standardized UIUC seeds (CS, ECE, Bioengineering, Crop Sciences, Information Sciences) and rates 15 shuffled, unlabeled cards (5 BridgeFinder, 5 Triadic Closure, 5 Topic-Only) on 7-point Likert scales for *Relevance*, *Novelty*, and *Outreach Actionability*, followed by the 10-item SUS questionnaire. Ratings are aggregated into per-participant means ($df = 14$) before running paired t-tests and Wilcoxon signed-rank tests with Bonferroni correction ($\alpha = 0.0167$).

Table 3: Project milestone schedule through final completion.

Target Date	Milestone	Status & Deliverables
Oct 30 (Done)	Mid-Semester	5k & 25k OpenAlex graphs, backtest, citation impact (1.128×), D3 prototype, Round 1 think-alouds ($N = 5$).
Nov 1–Nov 15	User Study R2	Run double-blind card study ($N = 15$) via study/index.html; participant-level paired t/Wilcoxon tests.
Nov 16–Nov 29	Iteration R3	Tune $(\alpha, \beta, \gamma, \lambda)$ trade-off against Topic-Only; SUS usability evaluation.
Dec 1–Dec 9	Final Delivery	Switch to native ACM acmart, 10–12 page final report, 10-min video demo.

7 Division of Labor & AI Disclosure

Who Did What: Evan Lin (elin26@illinois.edu): Implemented fetch_openalex.py (5k and 25k datasets), backtest.py (Louvain partitioning and baseline evaluation), and the D3.js v7 campus map in prototype/web/index.html. Eric Liu (ericl8@illinois.edu): Formulated the brokerage scoring model (Eq. (2)), implemented citation_impact.py, prepare_web_data.py, and the user study suite (generate_cards.py, study/index.html, analyze_study.py), and led the Round 1 student think-aloud sessions. Both partners co-authored this report.

Generative AI Disclosure: In compliance with CS 470 policy, we used Claude (Anthropic) as a collaborator to debug OpenAlex pagination, structure the D3 force layout, audit code-to-paper consistency, and refine LaTeX typesetting. Both authors independently verified all code, mathematical equations, and empirical measurements.

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